

JKBOSE 2026

CLASS 11TH

Plant Growth and Development

LECTURE-03

BOTANY

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Topics To Covered

- 1** Plant Growth Regulators
- 2** Auxin
- 3** Gibberlins → Next lecture.

Plant Growth Regulators (PGR's)



- ≡ AKA plant growth substances / plant hormones / phytohormones.
- ≡ These are small chemical compounds having diverse chemical nature.
- ≡ PGR's — divided into 02 broad categories:

a. plant growth promoters → involved in growth activities like cell elongation, cell division etc.
ex. Auxin, Gibberlins, Cytokinin.

b. plant growth inhibitors → involved in growth inhibiting activities like seed dormancy, leaf fall, fruit fall etc.
ex. Ethylene, Abscissic Acid (ABA).

Note

→ Ethylene acts as both $\left\{ \begin{array}{l} \text{Plant Growth promoter} \\ \text{plant Growth Inhibitor} \end{array} \right.$

→ Ethylene is largely - plant Growth Inhibitor.

AUXIN

- Term Auxin - derived from Greek word "Auxein" → which means to grow.
- Auxin - acidic in nature.
- Precursor: Amino acid (Tryptophan).

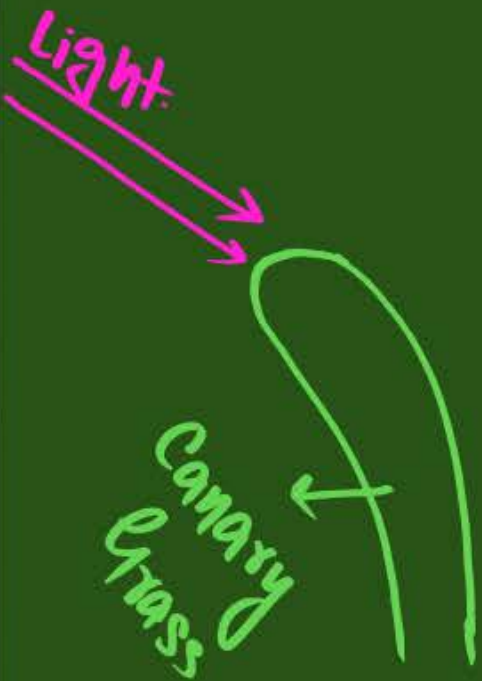
Discovery of Auxin

→ Charles Darwin & his son Francis Darwin

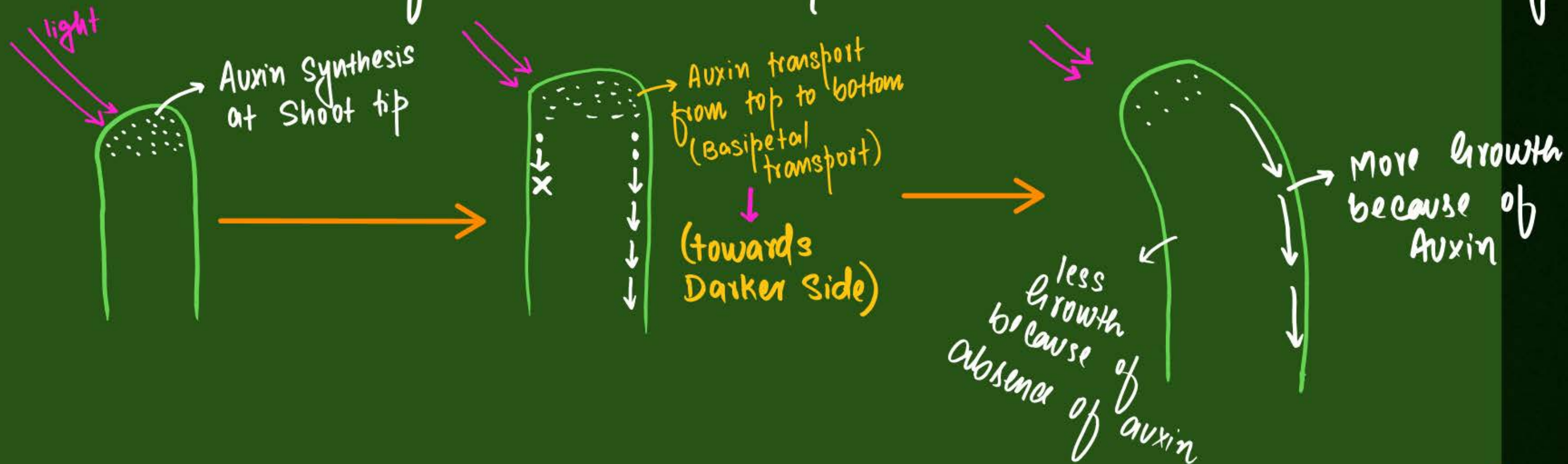
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worked on Canary Grass (monocot)

They observed - Shoot tip (i.e., coleoptile) of Canary grass shows bending towards light. (This is phototropism)

→ Charles Darwin & his son → Credited for discovery of phototropism.



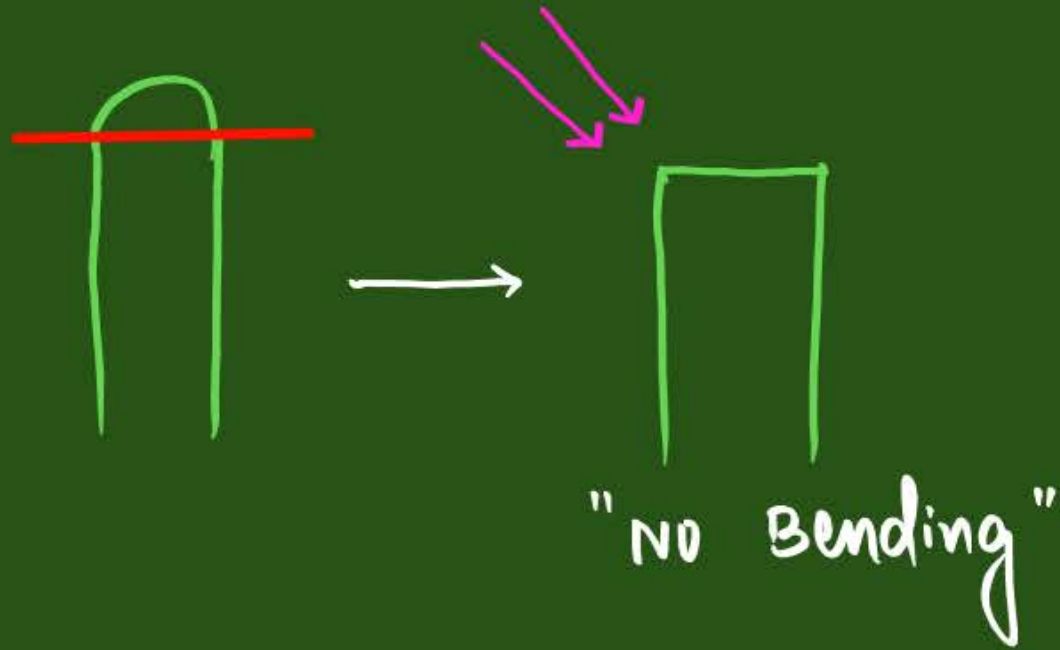
→ After a series of experiments, it was concluded that, shoot tip (coleoptile) was the site of transmittable influence that caused its bending.



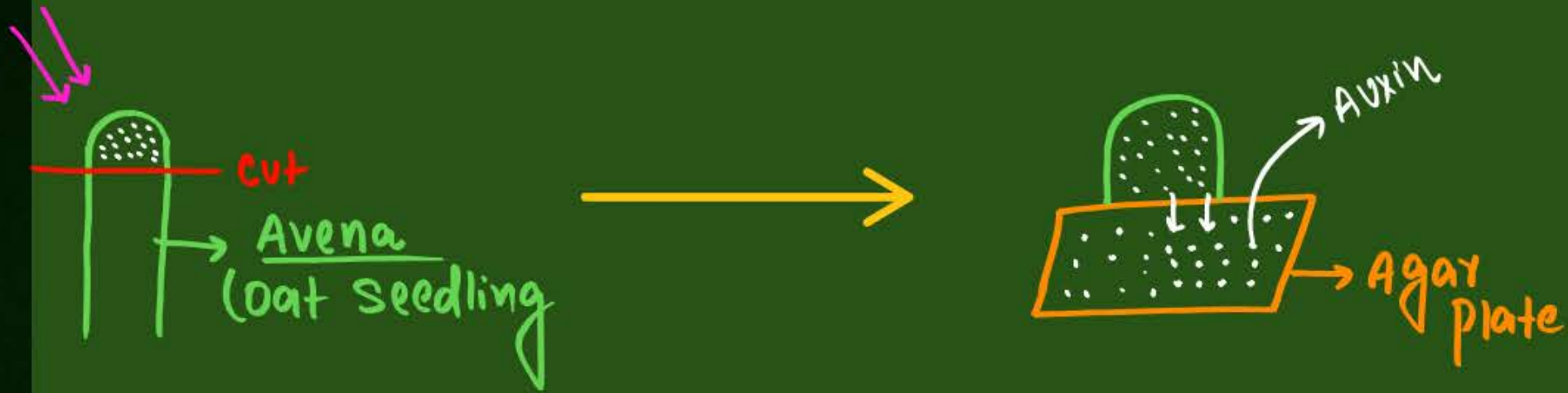
Due to this unequal growth shoot tip bends towards light.

→ Decapitation (ie, Removal of shoot tip).

↓
No bending of Shoot towards light.



→ **F. W. Went** → isolated Auxin from shoot tip of Avena (oat seedling) by using Agar-plate.



→ Term Auxin was coined by F. W. Went.

→ F. W. Went — credited for discovery of Auxin.

→ **Kogl Smith** — isolated Auxin from human urine.

Bioassay

It is a test used to determine:

- a. Function of hormone.
- b. Concentration of hormone at which it functions.

→ Bio-assay for Auxin → Avena Curvature Coleoptile test.

→ Auxin — two types:

a. Natural Auxin: produced naturally in plants and can be extracted.

Ex. IBA (indole-3-Butyric acid)
IAA (indole-3-Acetic acid)

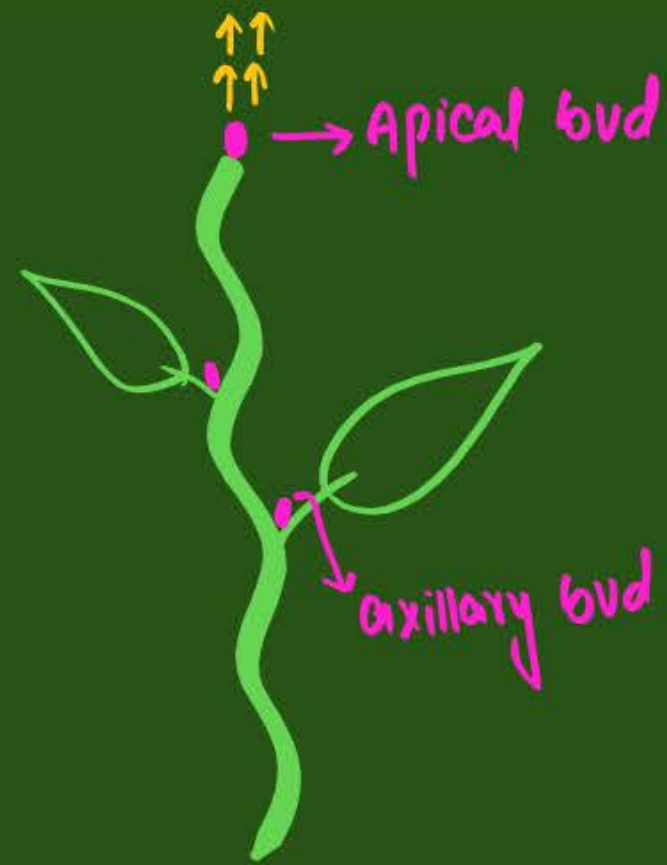
b. Synthetic Auxin: Man-made auxins.

Ex. NAA (Naphthalene acetic acid)

2,4D (2,4 Dichloro phenoxy acetic acid).

2,4,5T (2,4,5 Trichloro phenoxy acetic acid).

ROLE OF AUXIN



①. Auxin promotes apical dominance (growth of apical bud which inhibits the growth of axillary buds).

→ Decapitation (Removal) of Apical bud → It promotes growth of lateral buds i.e., apical dominance breaks.

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It is used in tea plantation and Hedge making.

② Auxin promotes flowering in pineapple.

③ Auxin promotes parthenocarpy (ex. in Tomato) — benefits to juice industry.

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Fruit formation
without seeds.

④ Auxin → promotes abscission in old leaves / fruits.

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Fall

→ prevents abscission in young leaves / fruits.

⑤ Auxin namely 2,4D → used by gardeners to make weed-free lawns.
It acts as weedicide and kills mature dicot weeds.

- ⑥. Auxin promotes xylem differentiation.
- ⑦. Auxin causes cell elongation.

GIBBERLINS

↓
Nex lechvre ??

QUESTIONS

The plant hormone used to destroy weeds in a field is

(NEET 2021)

- 1 IAA
- 2 NAA
- ✓ 3 2,4-D
- 4 IBA

HOMEWORK



Revision



Thank
You